

Vibrations Workshop 6 - MATLAB'n Adventures in the Electrical-Mechanical Analogy

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ME301

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General Overview

In this assignment, the objective is draw parallels between electrical and mechanical systems by analyzing Newton's Second Law and Kirchhoff's Voltage Law in the Laplace domain.

In particular, for all provided mechanical system, an equivalent circuit diagram will be illustrated. The respective equations of motion will be determined as well in the Laplace domain. In conjunction, using MATLAB, the coordinate time evolutions will be plotted using Laplace provided initial conditions.

If not clear, both Newton's Second Law and Kirchhoff's Voltage Law will be assumed throughout this analysis with the understanding that:

- Positively Oriented Force $F \longleftrightarrow$ Positively Oriented Voltage Source with Potential Difference F
- Spring with Stiffness $k \longleftrightarrow$ Capacitor with Capacitance $1/k$
- Mass $m \longleftrightarrow$ Inductor with Inductance m
- Damper with Damping Constant $R \longleftrightarrow$ Resistor with Resistance R

An important note is that all diagrams to follow were derived from the provided (illustrated) systems in the separate workshop pdf for this assignment.

Problem 1

The impedance circuit diagram is illustrated below. Note that the initial conditions (loosely denoted $I.C.$) will later be explicitly written out/encoded.

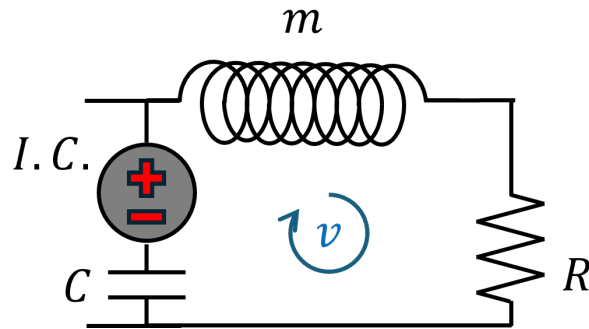


Figure 1: Equivalent Circuit 1

The equation of motion encoded by this circuit directly follows from Kirchhoff's Voltage Law via impedance (with Laplace parameter s and $\mathcal{L}\{v\} = V$):

$$\left(ms + R + \frac{1}{Cs} \right) V(s) = mv_0$$

Provided that δ is the Dirac Delta, here it is noted that the force drop across the force generator, which encodes the $I.C.$, is given by $\mathcal{L}^{-1}\{mv_0\}(t) = mv_0\delta(t)$. This forces $v(t=0) = v_0$

Now the velocity, position (using $\mathcal{L}\{x\}(s) = \mathcal{L}\{v\}(s)/s - x_0/s$), and admittance (using $Y(s) = (ms + R + 1/Cs)^{-1}$) will be plotted via MATLAB noting $m = 0.5$ kg, $C = 1.2665 \cdot 10^{-6}$ m/N.

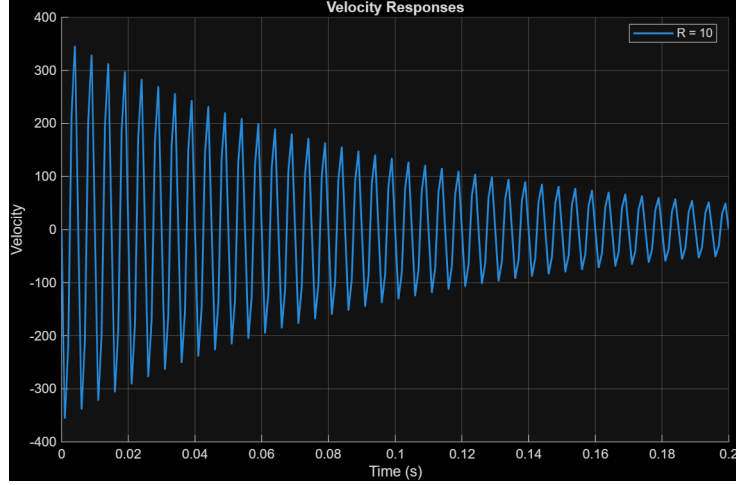


Figure 2: Velocity Evolution ($R = 10$ kg/s)

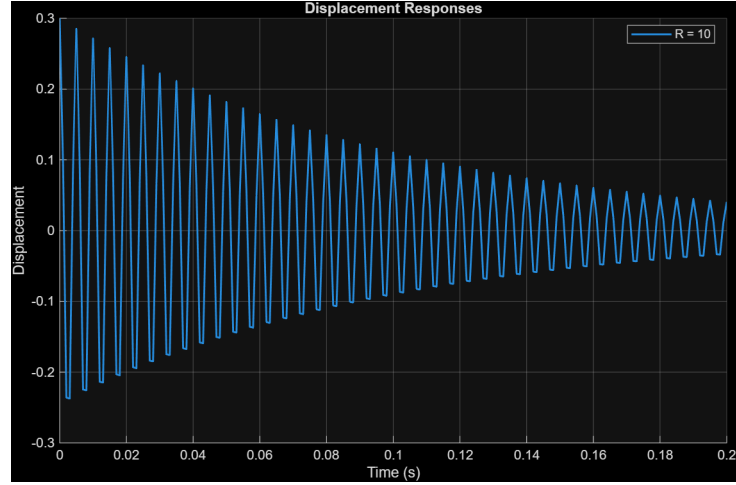


Figure 3: Position Evolution ($R = 10$ kg/s)

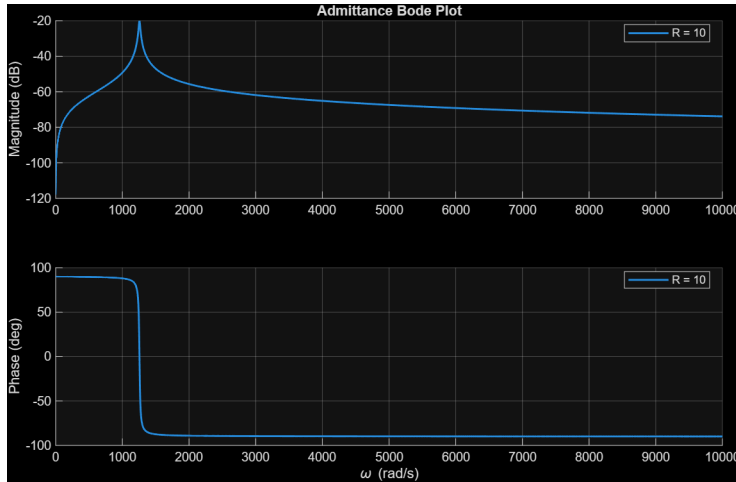


Figure 3: Admittance BODE Plot ($R = 10$ kg/s)

As great as those plots are, per the instruction, the dynamical responses will be plotted for various R values. The results are shown below (keeping all other parameters fixed):

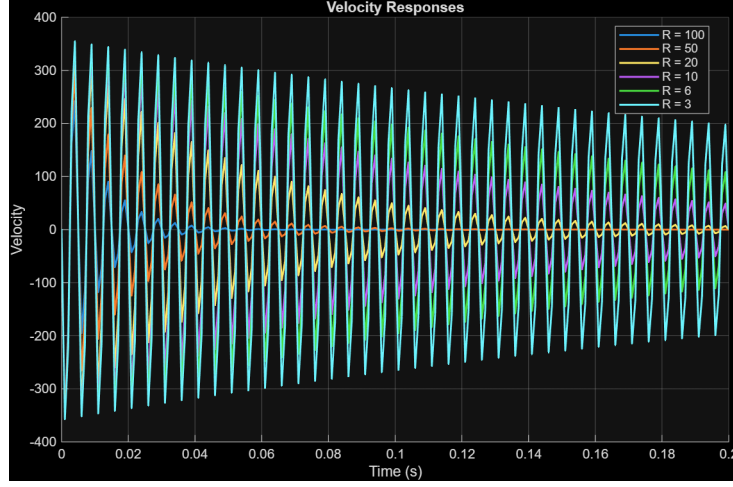


Figure 4: Velocity Evolutions $R = [100, 50, 20, 10, 6, 3]$ kg/s

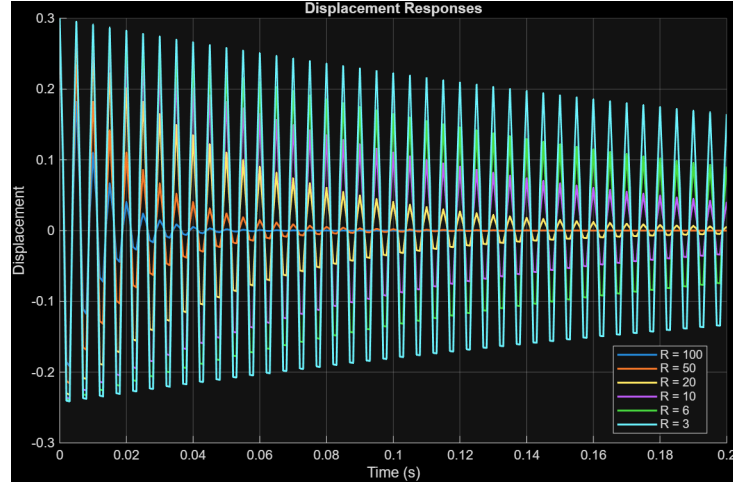


Figure 5: Position Evolutions $R = [100, 50, 20, 10, 6, 3]$ kg/s

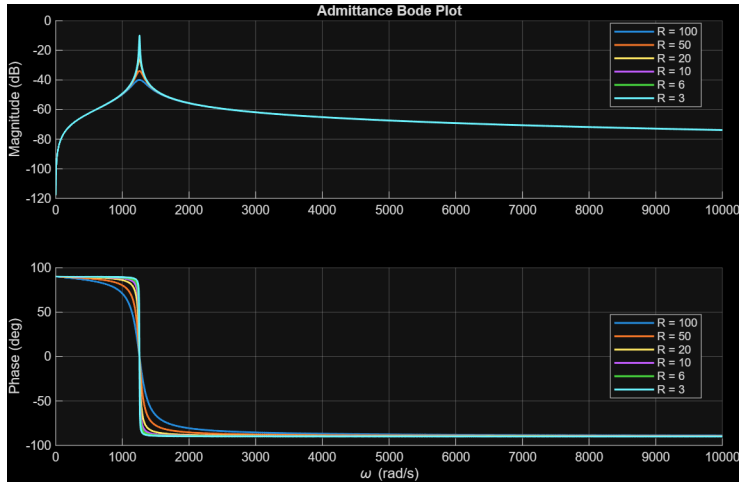


Figure 6: Admittances $R = [100, 50, 20, 10, 6, 3]$ kg/s

It can be observed that as R increases, the trajectories decay out faster, and so the peaks of the admittance plots go down. Likewise, as R decreases, the trajectories decay out slower, and so the peaks of the admittance plots go up. The same procedure will be carried out on the compliances (keeping everything else fixed).

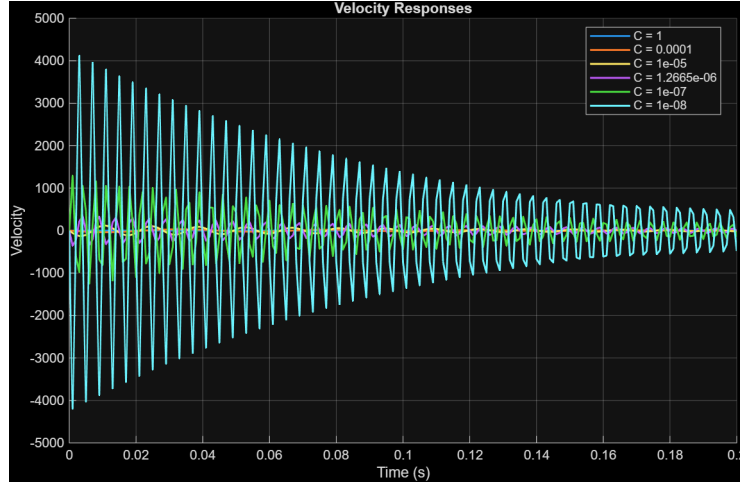


Figure 7: Velocity Evolutions $C = [1, 10^{-4}, 10^{-5}, 1.2665 \cdot 10^{-6}, 10^{-7}, 10^{-8}]$ m/N

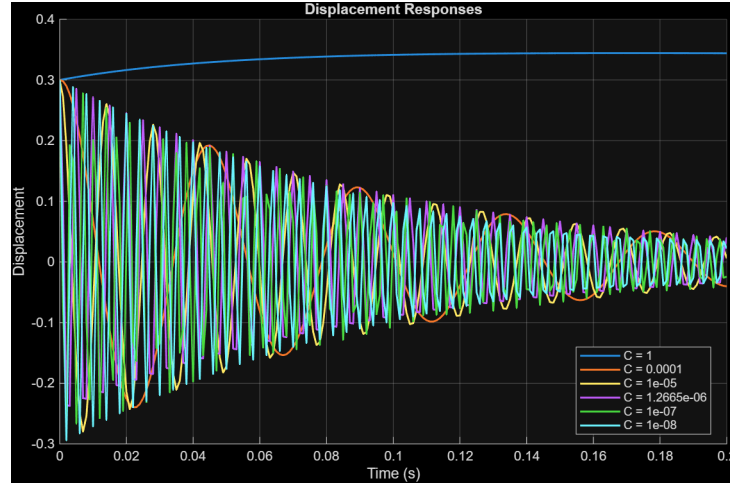


Figure 8: Position Evolutions $C = [1, 10^{-4}, 10^{-5}, 1.2665 \cdot 10^{-6}, 10^{-7}, 10^{-8}]$ m/N

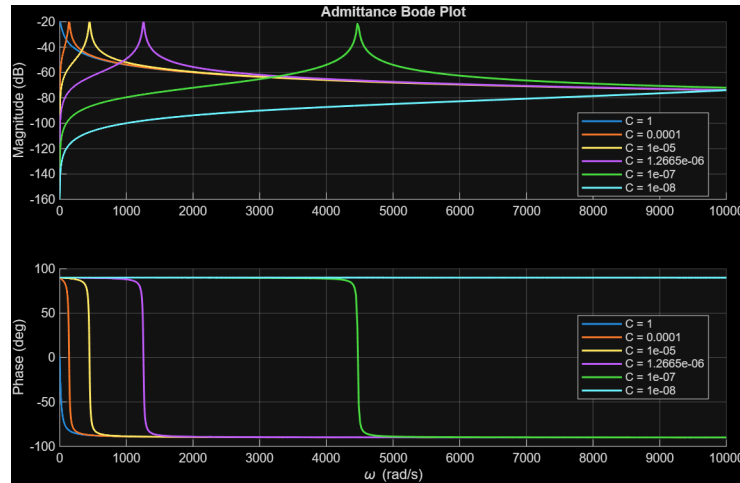


Figure 9: Admittances $C = [1, 10^{-4}, 10^{-5}, 1.2665 \cdot 10^{-6}, 10^{-7}, 10^{-8}]$ m/N

Here it can be seen that as compliance is changed, the trajectories (generally) decay out the same, and so the peaks of the admittance are at equal heights. However, increasing compliance reduces the oscillation frequency, and so the admittance peaks shift rightward as compliance decreases.

Problem 2

The impedance circuit diagram is illustrated below. Note that the initial conditions (loosely denoted *I.C.*) will later be explicitly written out/encoded.

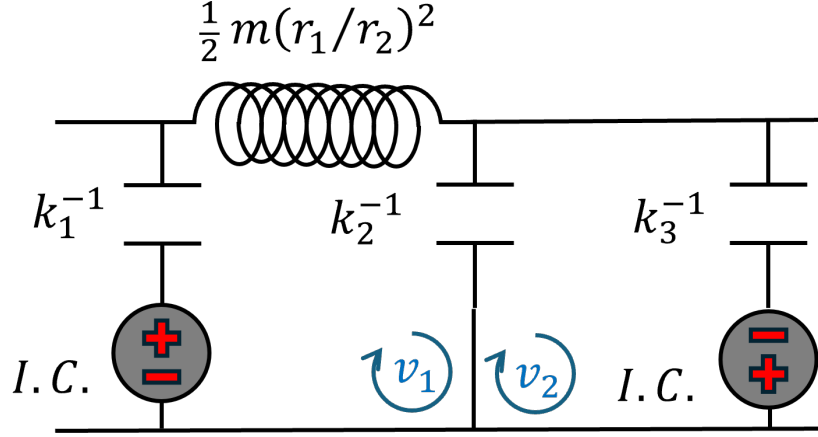


Figure 10: Equivalent Circuit 2

Note that the effective mass of the system is given by $m_{\text{eff}} = \frac{1}{2}m(r_1/r_2)^2$ precisely because the moment of inertia is $I = \frac{1}{2}mr_1^2$ and the mass is translated linearly at a radial distance of r_2 from the pivot point (for small angular displacements). This is all to say,

$$KE = \frac{1}{2}I(\dot{x}/r_2)^2 = \frac{1}{2}(I/r_2^2)\dot{x}^2 = \frac{1}{2}m_{\text{eff}}\dot{x}^2 \implies m_{\text{eff}} = I/r_2^2 = \left(\frac{1}{2}mr_1^2\right)/r_2^2 = \frac{1}{2}m(r_1/r_2)^2$$

With all of this in mind, the initial displacement of the mass is given by $x_{1_0} = r_2\theta_0$. At the massless node between the two springs, via a force balance:

$$\sum F = k_2(x_{1_0} - x_{2_0}) - k_3x_{2_0} = 0 \implies x_{2_0} = \frac{k_2}{k_2 + k_3}x_{1_0} = \frac{k_2}{k_2 + k_3}r_2\theta_0$$

All initial velocities will be assumed to be zero. Therefore, it immediately follows:

$$x_0 = \begin{bmatrix} x_{1_0} \\ x_{2_0} \end{bmatrix} = \begin{bmatrix} r_2\theta_0 \\ \frac{k_2}{k_2 + k_3}r_2\theta_0 \end{bmatrix} \quad \text{and} \quad v_0 = \begin{bmatrix} v_{1_0} \\ v_{2_0} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Now, the equations of motion encoded by this circuit directly follow from Kirchhoff's Voltage Law via impedance (with Laplace parameter s and $\mathcal{L}\{v_i\} = V_i$ for $i = 1, 2$):

$$\begin{bmatrix} \frac{1}{2}m(r_1/r_2)^2s + k_2/s + k_1/s & -k_2/s \\ -k_2/s & k_3/s \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \frac{-1}{s} \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix} \begin{bmatrix} x_{1_0} \\ x_{2_0} \end{bmatrix}$$

From this perspective it easy to see (more or less) that the force drop across the force generator in the first loop is $(k_1 + k_2)x_{1_0} - k_2x_{2_0}$. Likewise the force drop across the force generator in the second loop is $-k_2x_{1_0} + (k_2 + k_3)x_{2_0}$. If not clear as to where the RHS of the above matrix equation came from, noting the structure $ZV = -Kx_0/s$, one can derive the K matrix by computing $K = sZ(s)|_{s=0}$.

Noting that $m = 10$ kg, $k_1 = 200$ N/m, $k_2 = 300$ N/m, $k_3 = 500$ N/m, $r_1 = 0.5$ m, $r_2 = 0.3$ m, $\theta_0 = 0.15$ rad, the resultant positions, velocities, and admittances ($Y = Z^{-1}$) will be plotted.

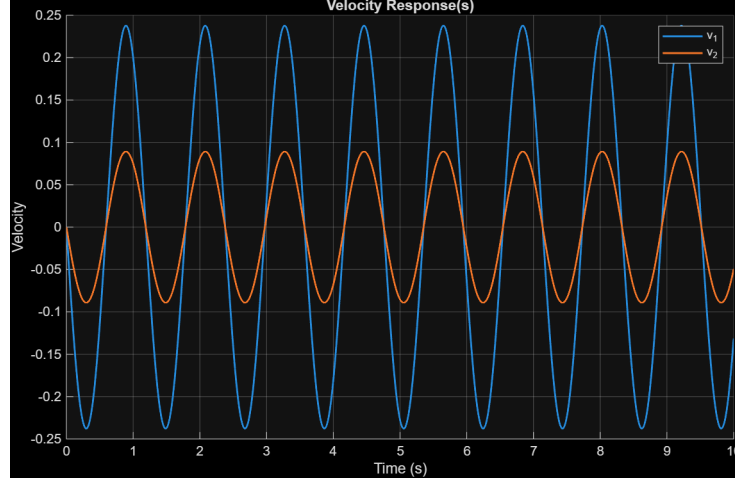


Figure 11: Velocity Responses

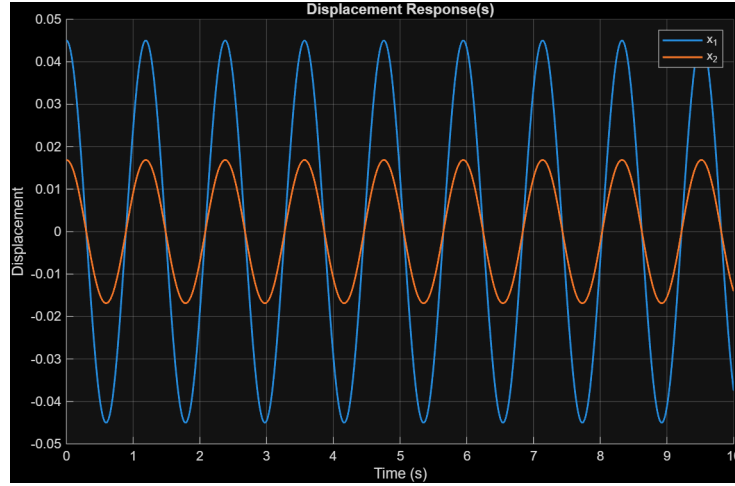


Figure 12: Position Responses

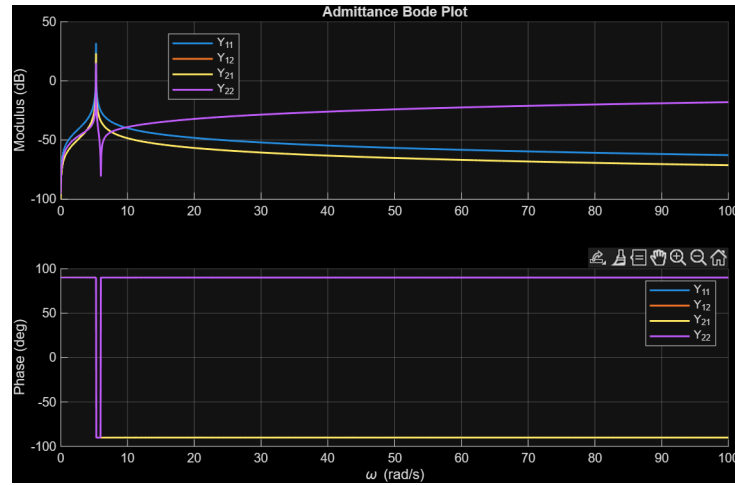


Figure 13: Admittance Responses

It will be explicitly mentioned that v_1 and x_1 correspond to the kinematics of the point (mass) on the rotating disc. Now, from the admittance plots, one can observe that the peaks are VERY sharp, which indicate low damping. In this case there is no damping.

Changing out the capacitor with capacitance $1/k_2$ with a resistor with resistance $R = 10 \text{ kg/s}$ modifies the (Laplaced) EOMs as follows with the previous understanding of the K matrix in conjunction with the omission of k_2 (i.e. $k_2 = 0$):

$$\begin{bmatrix} \frac{1}{2}m(r_1/r_2)^2s + R + k_1/s & -R \\ -R & k_3/s \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \frac{-1}{s} \begin{bmatrix} k_1 & 0 \\ 0 & k_3 \end{bmatrix} \begin{bmatrix} x_{1_0} \\ x_{2_0} \end{bmatrix}$$

Although the equivalent circuit was not explicitly drawn here, upon replacing the capacitor (as in figure 10) with the mentioned resistor, in that picture, the force drop across the force generator in the first loop is given by $k_1x_{1_0}$. Likewise, in the second loop, the force drop is $k_3x_{2_0}$.

The corresponding admittance responses will now be plotted via MATLAB as well just below:

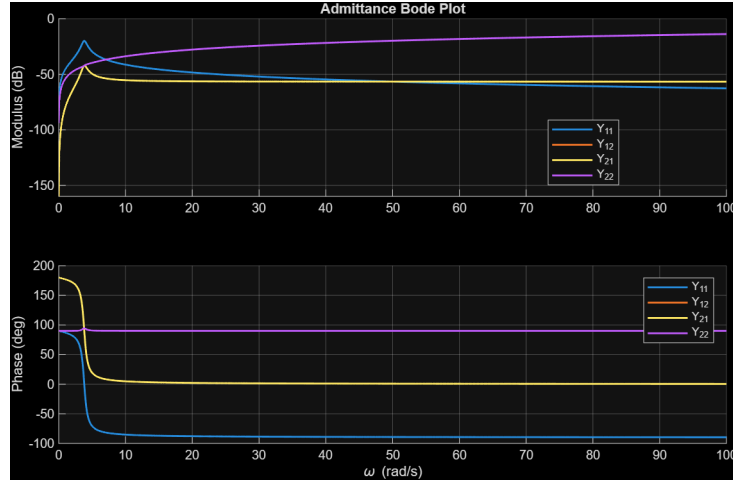


Figure 14: New Admittance Responses

One can see that the admittance peaks have softer peaks that also happen to occur at lower frequencies. This would imply there is more damping (or that there is damping to begin with), and that the resonant frequency of the system is lower.

Conclusionary Remarks

Working through this assignment brought light upon the ability to analyze involved mechanical systems in the Laplace domain. In particular, the admittance matrices for a given system entirely encode the dynamics of the system in conjunction with a set of initial conditions. In this sense, the admittance characterizes the behavior which ultimately can be used to predict the system's time evolution at a qualitative level by eye.

Matlab Code - Problem 1

```

1  clc; clear; close all;
2
3  % Parameters
4  m = 0.5;
5  % C = 1.2665e-6;
6  Cvals = [1 1e-4 1e-5 1.2665e-6 1e-7 1e-8];
7  R = 10;
8  % Rvals = [100 50 20 10 6 3];
9
10 % Time parameters
11 t0 = 0;
12 t1 = 0.2;
13 dt = 0.001;
14 t = t0:dt:t1;
15
16 % Laplace variable
17 s = tf('s');
18
19 % Frequency vector (extended to capture resonance)
20 w = logspace(0,4,2000);
21
22 % Initial Conditions
23 F = 0;
24 x0 = 0.3;
25 v0 = 1;
26
27 %% ===== Velocity Figure =====
28 figure(1); clf; hold on; grid on;
29 title('Velocity Responses')
30 xlabel('Time (s)')
31 ylabel('Velocity')
32
33 %% ===== Displacement Figure =====
34 figure(2); clf; hold on; grid on;
35 title('Displacement Responses')
36 xlabel('Time (s)')
37 ylabel('Displacement')
38
39 %% ===== Bode Figure =====
40 figure(3); clf;
41
42 subplot(2,1,1); hold on; grid on;
43 title('Admittance Bode Plot')
44 ylabel('Magnitude (dB)')
45
46 subplot(2,1,2); hold on; grid on;
47 ylabel('Phase (deg)')
48 xlabel('\omega (rad/s)')
49
50 legend_entries = cell(length(Cvals),1);
51
52 % for p = 1:length(Rvals)

```

```

53  for p = 1:length(Cvals)
54
55      % R = Rvals(p);
56      C = Cvals(p);
57
58      % Impedance (mechanical form)
59      Z_tf = m*s + R + 1/(C*s);
60
61      % Mechanical coefficients
62      M = m;
63      K = 1/C;
64
65      % Augmented force (Laplace domain)
66      F_aug = F + M*v0 - (K*x0)/s;
67
68      % Admittance
69      Y = inv(Z_tf);
70
71      % Laplace variables
72      V = Y*F_aug;
73      X = V/s + x0/s;
74
75      % Time responses
76      [v_resp, t_out] = impulse(V, t);
77      [x_resp, ~] = impulse(X, t);
78
79      v = squeeze(v_resp);
80      x = squeeze(x_resp);
81
82      %% Plot velocity
83      figure(1)
84      plot(t_out, v, 'LineWidth',1.4)
85
86      %% Plot displacement
87      figure(2)
88      plot(t_out, x, 'LineWidth',1.4)
89
90      %% Bode plots
91      [mag,phase] = bode(Y, w);
92      mag = squeeze(mag);
93      phase = squeeze(phase);
94
95      figure(3)
96      subplot(2,1,1)
97      semilogx(w,20*log10(mag), 'LineWidth',1.4)
98
99      subplot(2,1,2)
100     semilogx(w,phase, 'LineWidth',1.4)
101
102     % legend_entries{p} = sprintf('R = %g', R);
103     legend_entries{p} = sprintf('C = %g', C);
104
105 end
106

```

```
107 %% Apply legends
108
109 figure(1)
110 legend(legend_entries, 'Location', 'best')
111
112 figure(2)
113 legend(legend_entries, 'Location', 'best')
114
115 figure(3)
116 subplot(2,1,1)
117 legend(legend_entries, 'Location', 'best')
118
119 subplot(2,1,2)
120 legend(legend_entries, 'Location', 'best')
```

Matlab Code - Problem 2

```

1  clc; clear; close all;
2
3  %% Parameters
4  m = 10;
5  k1 = 200;
6  k2 = 300;
7  k3 = 500;
8  r1 = 0.5;
9  r2 = 0.3;
10 theta0 = 0.15;
11 R = 10;
12
13 %% Impedance Matrix
14 % Z = @(s) ...
15 %     [0.5*m*(r1/r2)^2*s+(k1+k2)/s -k2/s;
16 %     -k2/s (k2+k3)/s];
17 Z = @(s) ...
18     [0.5*m*(r1/r2)^2*s+R+k1/s -R;
19     -R k3/s];
20
21 %% Control System Toolbox Implementation
22 s = tf('s');
23 Z_tf = Z(s);
24
25 %% Laplaced External Force Vector & Initial Conditions
26 F = [0 0];
27 x0 = [r2*theta0 k2/(k2+k3)*r2*theta0];
28 v0 = [0 0];
29
30 %% Time Domain Parameters
31 t0 = 0;
32 t1 = 10;
33 dt = 0.001;
34
35 %% Coefficient Extraction
36 syms z
37 Z_sym = Z(z);
38
39 n = size(Z(1),1);
40 for i = 1:n
41     for j = 1:n
42         Zij = Z_sym(i,j);
43         M(i,j) = double(limit(Zij/z,z,+inf));
44         K(i,j) = double(limit(Zij*z,z,0));
45         R(i,j) = double(limit(Zij - M(i,j)*z - K(i,j)/z, z, 1));
46     end
47 end
48
49 disp('M: '); disp(M);
50 disp('R: '); disp(R);
51 disp('K: '); disp(K);
52

```

```

53 %% Augmented Force Vector
54 F_aug = F' + M*v0' - (K*x0')/s;
55
56 %% Dependent Laplace Variables
57 Y = inv(Z_tf);
58 V = Y*F_aug;
59 X = V/s + x0'/s;
60
61 %% Time-Domain
62 t = t0:dt:t1;
63 [v_resp, t_out] = impulse(V, t);
64 [x_resp, ~] = impulse(X, t);
65
66 % Extract signals
67 v = squeeze(v_resp(:, :, 1));
68 x = squeeze(x_resp(:, :, 1));
69
70 %% Plot Velocities
71 figure; hold on; grid on;
72 for k = 1:n
73     plot(t_out, v(:,k), 'LineWidth',1.4);
74 end
75 xlabel('Time (s)');
76 ylabel('Velocity');
77 title('Velocity Response(s)');
78 legend(arrayfun(@(k) sprintf('v_%d',k),1:n,'UniformOutput',false));
79
80 %% Plot Displacements
81 figure; hold on; grid on;
82 for k = 1:n
83     plot(t_out, x(:,k), 'LineWidth',1.4);
84 end
85 xlabel('Time (s)');
86 ylabel('Displacement');
87 title('Displacement Response(s)');
88 legend(arrayfun(@(k) sprintf('x_%d',k),1:n,'UniformOutput',false));
89
90 %% BODE Plot
91 w = logspace(-2,2,1000);
92
93 figure; hold on; grid on;
94 legend_entries = {};
95
96 for i = 1:n
97     for j = 1:n
98
99         [mag,phase] = bode(Y(i,j), w);
100
101         mag = squeeze(mag);
102         phase = squeeze(phase);
103
104         subplot(2,1,1); hold on; grid on;
105         semilogx(w,20*log10(mag), 'LineWidth',1.4);
106         ylabel('Modulus (dB)');

```

```

107         title('Admittance Bode Plot');
108
109         subplot(2,1,2); hold on; grid on;
110         semilogx(w,phase,'LineWidth',1.4);
111         ylabel('Phase (deg)');
112         xlabel('\omega (rad/s)');
113
114         legend_entries{end+1} = sprintf('Y_{%d%d}',i,j);
115     end
116 end
117
118 subplot(2,1,1); legend(legend_entries,'Location','best');
119 subplot(2,1,2); legend(legend_entries,'Location','best');

```